



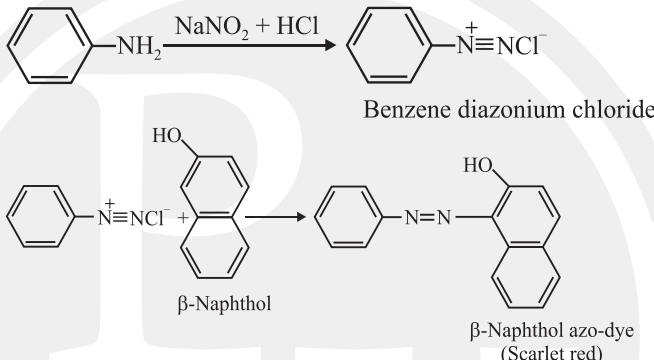
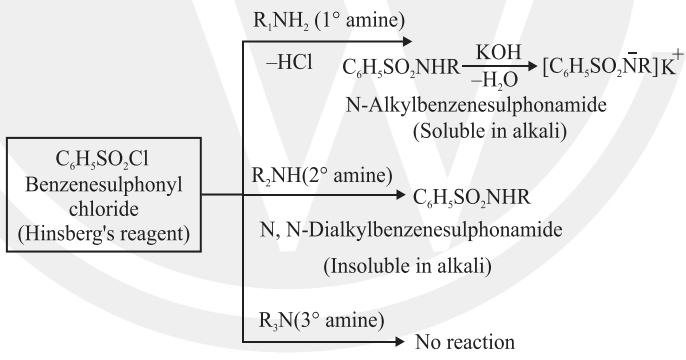
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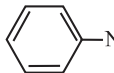
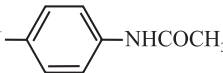
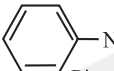
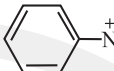
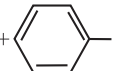
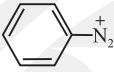
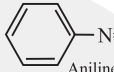
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Salt Analysis

Sr. No.	Concept	Description	Other information
1.	Detection of extra element		
	(a) Nitrogen	Lassaigne's extract (L.E.) $\text{Na} + \text{C} + \text{N} \xrightarrow{\Delta} \text{NaCN (L.E.)}$ $3\text{Na}_4[\text{Fe}(\text{CN})_6] + 4\text{FeCl}_3 \longrightarrow \text{Fe}_4[\text{Fe}(\text{CN})_6]_3 + 12\text{NaCl}$ prussian blue	L.E. + FeSO_4 + NaOH, boil and cool + FeCl_3 + conc. HCl Gives blue or green colour.
	(b) Sulphur	$2\text{Na} + \text{S} \longrightarrow \text{Na}_2\text{S (L.E.)}$ $\text{Na}_2\text{S} + \text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}] \longrightarrow \text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$ Sodium nitroprusside Deep violet $\text{Na}_2\text{S} + (\text{CH}_3\text{COO})_2\text{Pb} \xrightarrow{\text{CH}_3\text{COOH}} \text{PbS} \downarrow + 2\text{CH}_3\text{COONa}$ Black ppt.	L.E. + sodium nitroprusside. A deep violet colour. L.E. + CH_3COOH + $(\text{CH}_3\text{COO})_2\text{Pb}$ Gives a black ppt.
	(c) Halogens	$\text{Na} + \text{X} \xrightarrow{\Delta} \text{NaX (L.E.)}$ $\text{NaX} + \text{AgNO}_3 \xrightarrow{\text{HNO}_3} \text{AgX} \downarrow$ ppt. $\text{AgCl (White ppt.)} + 2\text{NH}_3(\text{aq.}) \longrightarrow [\text{Ag}(\text{NH}_3)_2]\text{Cl (Soluble)}$	L.E. + HNO_3 + AgNO_3. White ppt. soluble in aq. NH_3 (or NH_4OH) confirms Cl. Yellow ppt. partially soluble in aq. NH_3 (or NH_4OH) confirms Br. Yellow ppt. insoluble in aq. NH_3 or (NH_4OH) confirms I.

Sr. No.	Concept	Description	Other information
2.	Detection of functional group: (a) Hydroxyl group (alcoholic & phenolic) (i) Test of alcohol-Lucas test (ii) Test of phenols- - Ferric chloride test - Phthalein dye test.	$R-OH + HCl \xrightarrow[\text{Anhyd.}]{ZnCl_2} R-Cl + H_2O$ (Conc.) Alkyl chloride $6 \text{C}_6\text{H}_5\text{OH} + \text{FeCl}_3 \rightarrow 3\text{H}^+ + [\text{Fe}(\text{OC}_6\text{H}_5)_6]^{3-} + 3\text{HCl}$ (violet complex) (Colourless) (Pink) (b) Test of Carbonyl group (aldehyde & ketone)- (i) Test of aldehyde-Tollen's test (ii) Test of ketones-Iodoform test	Reagent = Anhyd. ZnCl_2 $1^\circ - \text{OH} \rightarrow$ No cloudiness $2^\circ - \text{OH} \rightarrow$ Cloudiness after 5 min $3^\circ - \text{OH} \rightarrow$ immediate cloudiness Violet complex is formed. Phenol condenses to give phenolphthalein which gives a dark pink colour with NaOH solution. Silver mirror is formed with aldehyde. Yellow precipitate of iodoform or triiodomethane.

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	(c) Carboxyl group	$\text{RCOOH} + \text{NaHCO}_3 \longrightarrow \text{RCOONa} + \text{H}_2\text{O} + \text{CO}_2$	Produce effervescence due to the formation of carbon dioxide.
	(d) Amino group in organic compound	$\text{RCOOH} + \text{R'OH} \xrightarrow[\Delta]{\text{Conc. H}_2\text{SO}_4} \text{RCOOR'} + \text{H}_2\text{O}$ Carboxylic acid Alcohol Ester	Carboxylic acid reacts with alcohols in the acidic medium to produce esters.
	(i) Carbylamine test	$\text{R-NH}_2 + \text{CHCl}_3 + 3\text{KOH} \xrightarrow{\Delta} \text{RNC} + 3\text{KCl} + 3\text{H}_2\text{O}$ (R = alkyl or aryl group) (Carbylamine)	Gave by aliphatic as well as aromatic primary amines.
	(ii) Azo dye test	 $\text{C}_6\text{H}_5\text{NH}_2 \xrightarrow{\text{NaNO}_2 + \text{HCl}} \text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$ (Benzene diazonium chloride) $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^- + \text{C}_{10}\text{H}_7\text{O} \longrightarrow \text{C}_6\text{H}_5\text{N}=\text{N}-\text{C}_{10}\text{H}_6\text{O}$ (β-Naphthol azo-dye (Scarlet red))	Forms β-naphthol to give a scarlet red dye, which is sparingly soluble in water.
	(iii) Hinsberg test	 C₆H₅SO₂Cl Benzenesulphonyl chloride (Hinsberg's reagent) R_1NH_2 (1° amine) $\xrightarrow{-\text{HCl}}$ $\text{C}_6\text{H}_5\text{SO}_2\text{NHR}$ $\xrightarrow[\text{-H}_2\text{O}]{\text{KOH}}$ $[\text{C}_6\text{H}_5\text{SO}_2\text{NR}]^-\text{K}^+$ N-Alkylbenzenesulphonamide (Soluble in alkali) R_2NH (2° amine) $\longrightarrow \text{C}_6\text{H}_5\text{SO}_2\text{NHR}$ N, N-Dialkylbenzenesulphonamide (Insoluble in alkali) R_3N (3° amine) $\longrightarrow$ No reaction	Reagent = HCl, KOH 1° amine – Soluble 2° amine – Insoluble 3° amine – No reaction
3.	Inorganic compounds:		
	(a) Mohr's salt	$\text{FeSO}_4 + (\text{NH}_4)_2\text{SO}_4 + 6\text{H}_2\text{O} \longrightarrow \text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ Ferrous ammonium sulphate (Mohr's salt)	Double salt is formed.
	(b) Potash alum	$\text{K}_2\text{SO}_4 + \text{Al}_2(\text{SO}_4)_3 + 24\text{H}_2\text{O} \longrightarrow \text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O} \text{ or } 2\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$ (Potash alum)	Double salt is formed.

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4.	Organic compounds: (a) Acetanilide (b) p-Nitro acetanilide (c) Aniline yellow	 Aniline $\xrightarrow[\text{Pyridine}]{\text{CH}_3\text{COCl}}$  Acetanilide $\text{HNO}_3 + 2\text{H}_2\text{SO}_4 \longrightarrow \text{NO}_2^+ + \text{H}_3\text{O}^+ + 2\text{HSO}_4^-$  Acetanilide $\xrightarrow[\text{Conc. H}_2\text{SO}_4]{\text{Conc. HNO}_3}$  p-nitroacetanilide (Major product)  Diazoaminobenzene $\xrightarrow{\text{HCl}}$  Benzenediazonium chloride  Aniline  Benzenediazonium chloride $\xrightarrow{\text{Aniline (in excess)}}$  Aniline yellow (p-Aminoazobenzene)	Mixture of the two acids releases nitronium ion(NO_2^+), which acts as an electrophile. Reaction is carried out only in weekly acidic conditions.																																
5.	Chemistry involved in titrimetric exercises: (a) Acid & Base Indicators (i) Acidic indicator (HIn) (ii) Basic indicator (InOH)	Phenolphthalein (HPh) $\text{HPh} \rightleftharpoons \text{H}^+ + \text{Ph}^-$ colourless pink Methyl orange (MeOH) $\text{MeOH} \rightleftharpoons \text{Me}^+ + \text{OH}^-$ yellow orange red <table><tr><th>Name of indicator</th><th>Colour in acidic medium</th><th>Colour in basic medium</th><th>Working pH range of indicators</th></tr><tr><td>Methyl orange (MeOH)</td><td>Orange red</td><td>Yellow</td><td>3.1 to 4.5</td></tr><tr><td>Methyl red</td><td>Red</td><td>Yellow</td><td>4.2 to 6.2</td></tr><tr><td>Phenol red</td><td>Yellow</td><td>Red</td><td>6.2 to 8.2</td></tr><tr><td>Phenolphthalein (HPh)</td><td>Colourless</td><td>Pink</td><td>8.2 to 10.2</td></tr></table> <table><tr><th>Type of titration</th><th>pH range of titration</th><th>Suitable indicators</th></tr><tr><td>SA/SB.</td><td>3 – 11</td><td>All indicators (MeOH, HPh etc.)</td></tr><tr><td>SA/WB</td><td>3 – 7</td><td>Methyl orange (MeOH)</td></tr><tr><td>WA/SB</td><td>7 – 11</td><td>Phenolphthalein (HPb)</td></tr></table>	Name of indicator	Colour in acidic medium	Colour in basic medium	Working pH range of indicators	Methyl orange (MeOH)	Orange red	Yellow	3.1 to 4.5	Methyl red	Red	Yellow	4.2 to 6.2	Phenol red	Yellow	Red	6.2 to 8.2	Phenolphthalein (HPh)	Colourless	Pink	8.2 to 10.2	Type of titration	pH range of titration	Suitable indicators	SA/SB.	3 – 11	All indicators (MeOH, HPh etc.)	SA/WB	3 – 7	Methyl orange (MeOH)	WA/SB	7 – 11	Phenolphthalein (HPb)	SA = Strong acid SB = Strong base WA = Weak acid WB = Weak base
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	(b) Oxalic acid vs KMnO_4	$5\text{H}_2\text{C}_2\text{O}_4 + 2\text{KMnO}_4 + 3\text{H}_2\text{SO}_4 \longrightarrow 10\text{CO}_2 + 2\text{MnSO}_4 + 8\text{H}_2\text{O} + \text{K}_2\text{SO}_4$	KMnO_4 act as oxidising agent. Oxalic acid acts as a reducing agent. The oxidation state of carbon in oxalic acid changes from +3 to +4, resulting in the production of carbon dioxide and the reduction of manganese in permanganate from +7 to +2.
	(c) Mohr's salt vs KMnO_4	$10(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O} + 2\text{KMnO}_4 + 8\text{H}_2\text{SO}_4 \longrightarrow 5\text{Fe}_2(\text{SO}_4)_3 + 2\text{MnSO}_4 + \text{K}_2\text{SO}_4 + 10(\text{NH}_4)_2\text{SO}_4 + 68\text{H}_2\text{O}$	Iron in Mohr's salt is oxidized from +2 to +3, while manganese in KMnO_4 is reduced from +7 to +2. The equivalence point is reached when all the iron ions are oxidized, and the purple color of KMnO_4 disappears.
6.	Qualitative analysis:		
	(a) Cations		
	(i) Tests of Pb^{2+}	$\text{PbCl}_2 + 2\text{KI} \longrightarrow \text{PbI}_2 + 2\text{KCl}$ (Hot solution) (Yellow precipitate)	
	(ii) Test for Copper ion $[\text{Cu}^{2+}]$	$3\text{CuS} + 8\text{HNO}_3 \longrightarrow 2\text{Cu}(\text{NO}_3)_2 + 2\text{NO} + 3\text{S} + 4\text{H}_2\text{O}$	
	(iii) Test for Aluminum ions $[\text{Al}^{3+}]$	$\text{AlCl}_3 + 3\text{NaOH} \longrightarrow \text{Al}(\text{OH})_3 + 3\text{NaCl}$ $\text{Al}(\text{OH})_3 + \text{NaOH} \longrightarrow \text{NaAlO}_2 + 2\text{H}_2\text{O}$ White gelatinous precipitate Sodium meta aluminate	
	(iv) Test for ferric ions $[\text{Fe}^{3+}]$	$\text{Fe}(\text{OH})_3 + 3\text{HCl} \longrightarrow \text{FeCl}_3 + 3\text{H}_2\text{O}$	

Sr. No.	Concept	Description	Other information
		$4\text{FeCl}_3 + 3\text{K}_4[\text{Fe}(\text{CN})_6] \longrightarrow \text{Fe}_4[\text{Fe}(\text{CN})_6]_3 + 12\text{KCl}$ Potassium ferrocyanide Prussian blue precipitate $\text{FeCl}_3 + \text{K}_4[\text{Fe}(\text{CN})_6] \longrightarrow \text{KFe}[\text{Fe}(\text{CN})_6] + 3\text{KCl}$ (Soluble prussian blue) $\text{Fe}^{3+} + \text{SCN}^- \longrightarrow [\text{Fe}(\text{SCN})]^{2+}$ (Blood red colour)	
	(v) Test for Zinc ion Zn^{2+}	$\text{ZnS} + 2\text{HCl} \longrightarrow \text{ZnCl}_2 + \text{H}_2\text{S}$ $\text{ZnCl}_2 + 2\text{NaOH} \longrightarrow \text{Zn}(\text{OH})_2 + 2\text{NaCl}$ $\text{Zn}(\text{OH})_2 + 2\text{NaOH} \longrightarrow \text{Na}_2\text{ZnO}_2 + 2\text{H}_2\text{O}$ $2\text{ZnCl}_2 + \text{K}_4[\text{Fe}(\text{CN})_6] \longrightarrow \text{Zn}_2[\text{Fe}(\text{CN})_6] + 4\text{KCl}$ Zinc ferrocyanide	
	(vi) Test for Nickel ion Ni^{2+}	$3\text{NiS} + 2\text{HNO}_3 + 6\text{HCl} \longrightarrow 3\text{NiCl}_2 + 2\text{NO} + 3\text{S} + 4\text{H}_2\text{O}$ Complex of red colour (Stable form of complex)	
	(vii) Test for Calcium ion $[\text{Ca}^{2+}]$	$\text{CaCO}_3 + 2\text{CH}_3\text{COOH} \longrightarrow (\text{CH}_3\text{COO})_2\text{Ca} + \text{H}_2\text{O} + \text{CO}_2$ $(\text{CH}_3\text{COO})_2\text{Ca} + (\text{NH}_4)_2\text{C}_2\text{O}_4 \longrightarrow (\text{COO})_2\text{Ca} + 2\text{CH}_3\text{COONH}_4$ Ammonium oxalate Calcium oxalate (White precipitate)	
	(viii) Test for Barium ion $[\text{Ba}^{2+}]$	$\text{BaCO}_3 + 2\text{CH}_3\text{COOH} \longrightarrow (\text{CH}_3\text{COO})_2\text{Ba} + \text{H}_2\text{O} + \text{CO}_2$ $(\text{CH}_3\text{COO})_2\text{Ba} + \text{K}_2\text{CrO}_4 \longrightarrow \text{BaCrO}_4 + 2\text{CH}_3\text{COOK}$ Barium chromate (yellow precipitate)	
	(ix) Test for Magnesium ion Mg^{2+}	$\text{NH}_4^+ + \text{CO}_3^{2-} \rightleftharpoons \text{NH}_3 + \text{HCO}_3^-$ $\text{Mg}^{2+} + \text{Na}_2\text{HPO}_4 \longrightarrow \text{Mg}(\text{NH}_4)\text{PO}_4 + \text{NH}_4\text{OH} + 2\text{Na}^+ + \text{H}_2\text{O}$ Magnesium ammonium phosphate (White precipitate)	
	(x) Tests for NH_4^+	$(\text{NH}_4)_2\text{SO}_4 + 2\text{NaOH} \longrightarrow \text{Na}_2\text{SO}_4 + 2\text{NH}_3 + 2\text{H}_2\text{O}$ $\text{NH}_2 + \text{HCl} \longrightarrow \text{NH}_4\text{Cl}$	

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	(b) Anions-		
	(i) Test of Carbonate ion $[\text{CO}_3^{2-}]$	$\text{Na}_2\text{CO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O} + \text{CO}_2$ $\text{Ca}(\text{OH})_2 + \text{CO}_2 \longrightarrow \text{CaCO}_3 + \text{H}_2\text{O}$	
	(ii) Test for Sulphide ion $[\text{S}^{2-}]$	$\text{CaCO}_3 + \text{CO}_2 + \text{H}_2\text{O} \longrightarrow \text{Ca}(\text{HCO}_3)_2$ $\text{Na}_2\text{S} + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{S}$ $(\text{CH}_3\text{COO})_2\text{Pb} + \text{H}_2\text{S} \longrightarrow \underset{\substack{\text{Lead sulphide} \\ \text{Black precipitate}}}{\text{PbS}} + 2\text{CH}_3\text{COOH}$	
	(iii) Test of Sulphate ions $[\text{SO}_4^{2-}]$	$\text{Na}_2\text{S} + \text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}] \longrightarrow \text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$ Sodium nitroprusside Complex of purple colour	
	(iv) Test for Nitrate ion $[\text{NO}_3^-]$	$\text{Na}_2\text{SO}_4 + \text{BaCl}_2 \longrightarrow \underset{\substack{\text{Barium sulphate} \\ \text{(White precipitate)}}}{\text{BaSO}_4} + 2\text{NaCl}$ $\text{NaNO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{NaHSO}_4 + \text{HNO}_3$ $4\text{HNO}_3 \longrightarrow 4\text{NO}_2 + \text{O}_2 + 2\text{H}_2\text{O}$ $2\text{NaNO}_3 + 4\text{H}_2\text{SO}_3 + 3\text{Cu} \longrightarrow \underset{\substack{\text{Copper sulphate} \\ \text{(Blue)}}}{3\text{CuSO}_4} + \text{Na}_2\text{SO}_4 + 4\text{H}_2\text{O} + 2\text{NO}$ $2\text{NO} + \text{O}_2 \longrightarrow \underset{\substack{\text{(Brown fumes)}}}{2\text{NO}_2}$	
	(v) Test for Nitrite ion $[\text{NO}_2^-]$	$2\text{NaNO}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + 2\text{HNO}_2$ $3\text{HNO}_2 \longrightarrow \text{HNO}_3 + 2\text{NO} + \text{H}_2\text{O}$ $2\text{NO} + \text{O}_2 \longrightarrow \underset{\substack{\text{Brown gas}}}{2\text{NO}_2}$ $\text{NO}_2^- + \text{CH}_3\text{COOH} \longrightarrow \text{HNO}_2 + \text{CH}_3\text{COO}^-$ $2\text{HNO}_2 + 2\text{KI} + 2\text{CH}_3\text{COOH} \longrightarrow 2\text{CH}_3\text{COOK} + 2\text{H}_2\text{O} + 2\text{NO} + \text{I}_2$ $\text{I}_2 + \text{Starch} \longrightarrow \text{Blue complex}$	
	(vi) Test for Chloride ion $[\text{Cl}^-]$	$\text{NaCl} + \text{H}_2\text{SO}_4 \longrightarrow \underset{\substack{\text{Sodium hydrogen} \\ \text{sulphate}}}{\text{NaHSO}_4} + \underset{\substack{\text{Hydrogen chloride} \\ \text{Colourless gas}}}{\text{HCl}}$ $\text{HCl} + \text{NH}_3 \longrightarrow \underset{\substack{\text{Ammonium chloride} \\ \text{white fumes}}}{\text{NH}_4\text{Cl}}$	

Sr. No.	Concept	Description	Other information
	(vii) Test for Bromide ion $[\text{Br}^-]$ (viii) Test for Iodide ion $[\text{I}^-]$	$2\text{NaBr} + 2\text{H}_2\text{SO}_4 \longrightarrow \text{Br}_2 + \text{SO}_2 + \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$ $2\text{NaBr} + \text{MnO}_2 + 2\text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + \text{MnSO}_4 + 2\text{H}_2\text{O} + \text{Br}_2$ $2\text{NaI} + 2\text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + \text{SO}_2 + 2\text{H}_2\text{O} + \text{I}_2$ $\text{I}_2 + \text{Starch solution} \longrightarrow \text{Blue colour}$ $\text{NaI} + \text{H}_2\text{SO}_4 \longrightarrow \text{NaHSO}_4 + \text{HI}$ $2\text{HI} + \text{H}_2\text{SO}_4 \longrightarrow 2\text{H}_2\text{O} + \text{I}_2 + \text{SO}_2$ $6\text{NaI} + 4\text{H}_2\text{SO}_4 \longrightarrow 3\text{I}_2 + 4\text{H}_2\text{O} + \text{S} + 3\text{Na}_2\text{SO}_4$ $8\text{NaI} + 5\text{H}_2\text{SO}_4 \longrightarrow 4\text{I}_2 + \text{H}_2\text{S} + 4\text{Na}_2\text{SO}_4 + 4\text{H}_2\text{O}$	
7.	Chemical principles involved in the following experiments: (a) Enthalpy of solution of CuSO_4 (b) Enthalpy of neutralization of strong acid & strong base (c) Preparation of lyophilic & lyophobic sols (d) Kinetic study of the reaction of iodide ions (I^-) with hydrogen peroxide (H_2O_2) at room temperature	$\text{CuSO}_4 (\text{s}) \rightarrow \text{CuSO}_4 (\text{aq.}); \Delta H_{(\text{sol.})}^0$ $\text{H}^+ (\text{aq}) + \text{OH}^- (\text{aq}) \longrightarrow \text{H}_2\text{O} (\text{l}), \Delta_{\text{neut}} H \text{ is negative}$ (Acid) (Base) These are colloidal solutions where the dispersed phase (colloid) has a strong affinity for the dispersion medium (solvent). The preparation of lyophilic sols typically involves solvating the colloid with the solvent, leading to stable colloidal solutions. In contrast, lyophobic sols are colloidal solutions where the dispersed phase has little affinity for the dispersion medium. Preparing lyophobic sols often involves vigorous dispersion methods, such as mechanical agitation or chemical treatments, to disperse the colloidal particles evenly in the medium.	$\Delta H_{(\text{sol.})}^0 =$ Enthalpy of solution $\Delta_{\text{neut}} H$ is known as enthalpy of neutralization.

Sr. No.	Concept	Description	Other information
	- Rate of reaction - Reaction mechanism - Catalysis	Rate of collision between iodide ions(I^-) and hydrogen peroxide (H_2O_2) molecules. Understanding the reaction mechanism, including any intermediate steps, is crucial. The kinetic study helps to identify reaction orders and RDS. In some cases, a catalyst (e.g. , iodide ion in this reaction) may be involved. Role of catalyst in lowering the activation energy & increasing the reaction rate can be explored.	

